



REVIEW PAPER

Strategic optimization of forest residues to bioenergy and biofuel supply chain

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SUMMARY

Forest residues are renewable materials for bioenergy conversion that have the potential to replace fossil fuels beyond electricity and heat generation. A challenge hindering the intensified use of forest residues for energy production is the high cost of their supply chain. Previous studies on optimal design of forest residue supply chains focused on biofuel or bioenergy production separately, mostly with a single time period approach. We present a multi-period mixed integer linear programming model that optimizes the supply chain of forest residues for the production of bioenergy and biofuels simultaneously. The model determines (i) the location, type and size of the technologies to install and the period to install them, (ii) the mix of biofuel and bioenergy products to generate, (iii) the type and amount of forest residues to acquire and the sourcing points, (iv) the amount of forest residues to transport from sources to facilities and (v) the amount of product to transport from facilities to markets. The objective of the model is to maximize the net present value of the supply chain over a 20-year planning horizon with yearly time steps. We applied the model to a case study in British Columbia, Canada, to investigate the production of heat, electricity, pellets and pyrolysis bio-oil from available forest harvesting residues and sawmill wastes. Based on current energy generation costs in the region and the predicted operating costs of new conversion plants, the results of our model recommended the installation of small biomass boilers coupled with steam turbines for electricity production (0.5 and 5 MW) and pyrolysis plants with a capacity of 200 and 400 odmt day⁻¹. We performed a sensitivity analysis to evaluate the sensitivity of the optimal result to changes in the demand and price of products, as well as the availability and cost of forest residues. Copyright © 2014 John Wiley & Sons, Ltd.

KEY WORDS

biofuels; bioenergy; forest residues; optimization; supply chain; mathematical programming; mixed integer programming

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1. INTRODUCTION

The use of forest residues for the production of bioenergy and biofuels has gained interest in recent years because it has the potential to generate additional revenue for the forest sector while reducing greenhouse gas emissions and helping forest-rich communities diversify their energy sources and achieve energy independence [1]. Forest residues include harvesting residues generated as by-products of conventional forest operations (tops, branches and non-merchantable logs) and sawmill wastes (chips, sawdust, shavings and bark) [2]. These residues

can be directly converted into bioenergy (e.g., heat and electricity), or processed into biofuels (e.g., ethanol, biodiesel and bio-gasoline) or other bio-based products (e.g., chemicals, plastics and other materials) [3].

Canada has a developed forestry sector that produces a large amount of forest residues [4]. Currently, mill wastes are mostly committed to other forest product mills (pellets and pulp) or used as an energy source within the mills [5]. However, there is an increasing interest in the use of harvesting residues for the production of energy [4]. The estimated amount of forest harvesting residues in Canada is 20 ± 0.6 million oven-dry metric tonnes (odmt) per year,

of which 42% is in British Columbia [6]. In addition, British Columbia could have a temporary resource of up to 11 million odmt per year of mountain pine beetle-killed wood until the year 2025 [7].

Currently, the total installed pellet capacity in Canada is about 3 million tonnes per year [8], and there is an installed bioenergy capacity of over 4800 MW [4]. Most of the installed capacity is based on direct combustion of mill wastes for heat production and/or steam for power generation in forest product mills [9]. Current interest in transforming the Canadian forest products industry motivated a recent study called the Biopathways Project [10] that estimated the potential return on capital employed for traditional and emerging bio-technologies for the production of bioenergy, biofuels and other bioproducts using forest residues. The study found that pyrolysis and combined heat and power (CHP) via gasification were the two most promising technologies in British Columbia, while pellet and torrefied pellet production were the most promising ones in Ontario, and pyrolysis and full fractionation (for the production of ethanol, furfural and lignin) in Quebec [10]. Research and demonstration projects developing combustion, gasification, pyrolysis, thermal and catalytic conversion, and fermentation technologies are underway for the production of bioenergy and biofuels from forest residues in different locations in Canada [9].

Despite the high interest and potential to produce bioenergy and biofuels from forest residues, several operational and economic challenges hinder their utilization. These challenges are related to capital investment, feedstock availability, quality and cost [11]. Because capital costs of energy producing technologies are high (e.g., in the order of \$490 to \$7400 per installed kW_e in combined heat and power plants depending on the type of technology and scale [12]), the economics of new projects rely heavily on achieving economies of scale [13]. This results in high demand for forest residues whose availability is linked with conventional harvesting operations and the production of traditional forest products. Typically, harvesting residues are distributed over vast regions, and their availability varies over time [11]. Different types of biomass have different quality attributes [14] that impact their procurement and preprocessing costs, as well as their transportation and conversion efficiencies [15]. In addition, forest biomass has lower energy density than a large number of fossil fuels [16], which results in the need for collection, processing, handling and transportation of large amounts of forest biomass over the service life of the conversion plant. This stresses the need for an optimal design of the supply chain [13].

Mathematical modeling and optimization have been used to support the planning and management of forest bioenergy supply chains [11]. For the production of heat and power utilizing forest biomass, a group of studies developed supply chain optimization models that considered forest biomass geographical distribution to address sourcing, capacity [17] and plant location decisions [18–20]. A large set of studies focused on the design of biofuel supply

chains mainly for bioethanol production from different types of agricultural biomass (e.g., [21–24]). These studies relied on annual biomass production (cultivation and harvesting) rates per supply area. Many biofuel supply chain studies that included forest residues in the feedstock mix relied on average annual production rates and average costs of forest residues per supply area [25–32]. This was also the case for two studies that modeled the combined production of bioenergy and biofuels [33,34]. Only a few papers considered the impact of temporal variation of parameters on strategic level decisions. In the biofuel supply chain models presented by Natarajan *et al.* [27], You *et al.* [32], Huang *et al.* [35] and Leduc *et al.* [36], the determination of when to open new biorefineries was based on biofuel demand variation. However, these models considered a constant annual supply of forest residues from each source over the life time of the project, and a constant forest residue cost. In the model developed by Feng *et al.* [37], an integrated bio-refinery and forest products model was proposed that permitted variability in the biomass supply over time, but the installation of plants was allowed only at time zero.

We developed a supply chain optimization model for the integrated production of bioenergy and biofuels from various types of forest residues and sawmill wastes (with different quality attributes and costs) through the installation of different conversion technologies. Our model was developed considering the variability per period of (i) available biomass quantities and cost per supply source, (ii) demand and price values of biofuel and bioenergy products, and (iii) transportation and operating costs. This is necessary because biomass supply (availability and cost) and market conditions (demand for and price of product) may vary over time. These variations should be considered in making strategic decisions such as the installation period of each selected plant, and tactical decisions such as the flow of biomass and products per period. The development of this model was motivated by a significant potential to utilize large quantities of forest biomass in the Williams Lake Timber Supply Area (TSA) located in the Interior Region of British Columbia, Canada. The objective function of the optimization model was to maximize the net present value (NPV) of a supply chain designed for the production of bioenergy and biofuels (heat, electricity, pellets and pyrolysis bio-oil) from available forest residues and sawmill wastes.

2. THE SUPPLY CHAIN OF FOREST RESIDUES TO BIOENERGY AND BIOFUELS

There are different factors that affect the overall profit of forest residues to bioenergy and biofuel supply chains: the sources and types of forest residues (harvesting residues and sawmill wastes), the location of plants, the type and capacity of technologies, the types of products to

produce, and the locations (markets) where they should be sold. In addition, optimal decisions on the amounts of biomass collected, transported and converted, and the quantity of bioenergy and biofuels produced and transported within the supply chain (material flows) per period are required. Also, when variations in feedstock supply or market conditions are expected, the installation period of plants must be determined. Figure 1 shows a general diagram of the supply chain modeled in this paper.

Harvesting residues include tree tops, branches and some small diameter logs and low quality non-merchantable logs (not suitable for either lumber or pulp production). Sawmill wastes encompass wood chips, sawdust, shavings, bark and hog fuel. Different wood species and biomass types (e.g., wood chips, bark, sawdust and shavings) have different characteristics (energy content, moisture content, particle size, ash and contaminant content [14,16,38]) that impact the economics of a bioenergy and biofuel project.

Typically, harvesting residues are scattered over large areas, aggregated in piles at the roadside of harvesting blocks in the forest and their procurement requires allocation of labor and equipment for collection, processing, drying and transportation to conversion facilities. The availability and cost of mill wastes depend on competing users (e.g., pulp mills, pellet mills, or other bioenergy and biofuel producers). The close proximity of conversion facilities to large sources of forest residues is key to reducing the costs of feedstock handling and transportation [38].

Biomass can be converted into heat, electricity and biofuels such as bio-oil and ethanol [16]. The selection of conversion processes and technologies is determined by the type and quantity of available biomass, and the desired form of energy [39]. Three basic process categories exist for the conversion of biomass into bioenergy and biofuels, namely, biochemical, chemical and thermochemical [39,40]. Biochemical processes (such as ethanol production through hydrolysis and fermentation) and thermochemical processes (such as combustion, pyrolysis and gasification) are the most suitable ones to convert forest biomass into bioenergy and

biofuels [40]. Densification processes are used to convert forest residues into merchantable products such as pellets, briquettes and wood chips [41].

Typically, bioenergy supply chains cover activities up to the heat and/or power plant, while biofuel supply chains include biofuel distribution to markets or distribution terminals.

3. MATHEMATICAL MODEL

A multi-period mixed integer linear program (MILP) model is developed in this paper to optimize the strategic design of the supply chain of forest residues to bioenergy and biofuels. Table I presents the indices, sets, parameters and decision variables of the model.

The mathematical formulation of the objective function is presented in Equation (1):

$$\text{Max NPV} = \sum_t \frac{1}{(1 + \alpha)^{t-1}} (\text{TotalRevenue}_t - \text{TotalCost}_t) \quad (1)$$

The total revenue in period t is determined by the selling price per unit of each biofuel or bioenergy product multiplied by the amount sold during period t (Eq. (2)):

$$\text{TotalRevenue}_t = \sum_{nm} (\text{BiofuelPrice}_{nmt} * \sum_j w_{njmt}) + \sum_{oj} (\text{BioenergyPrice}_{ojt} * w_{ojt}) \quad (2)$$

The total cost, in each period, includes the purchase cost of biomass, the transportation cost of biomass, fixed costs associated with each technology (annualized capital investment, salaries and maintenance costs), variable costs (feedstock dependent costs such as wages and operational costs) and the transportation cost of products to markets during period t (Eq. (3)).

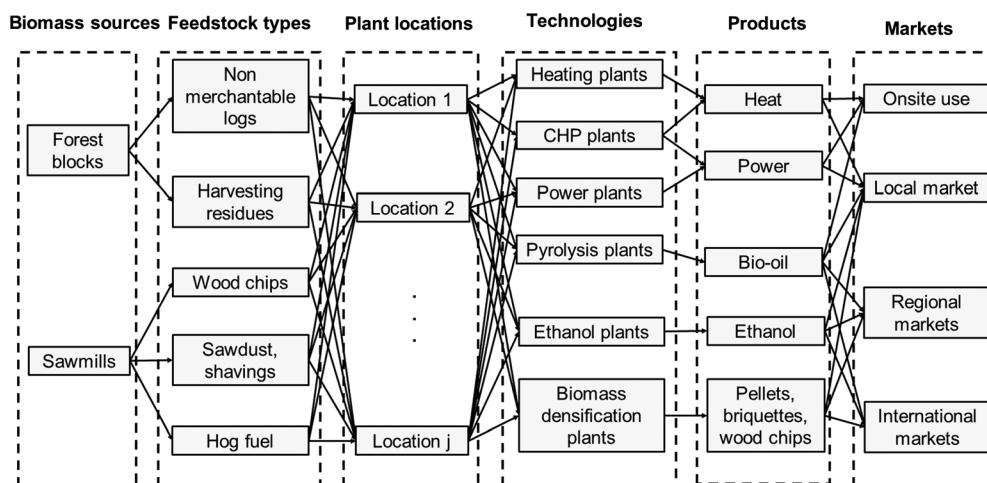


Figure 1. Supply chain of forest residues to bioenergy and biofuel. CHP, combined heat and power.

Table I. List of indices, sets, parameters and decision variables of the model.

Element	Description
Sets/Indices	
$F = \{1, 2, \dots, f\}$	Set of biomass feedstock types (e.g., non-merchantable logs, harvesting residues, sawdust and shavings, and hog fuel)
$I = \{1, 2, \dots, i\}$	Set of candidate biomass sources (e.g., forest blocks and mills)
$J = \{1, 2, \dots, j\}$	Set of candidate locations for bioenergy and biofuel plants
$L = \{1, 2, \dots, l\}$	Set of candidate biomass conversion technologies
$M = \{1, 2, \dots, m\}$	Set of candidate markets (for biofuels)
$N = \{1, 2, \dots, n\}$	Set of biofuels that are transported to a market (e.g., bio-oil and densified products)
$O = \{1, 2, \dots, o\}$	Set of bioenergy products that are sold at the plant location (e.g., heat and electricity)
$T = \{1, 2, \dots, t\}$	Time periods (e.g., years)
Parameters	
α	Depreciation rate
AvailableBiomass _{fit}	Available biomass type f at source i in period t (odmt ^a)
BioenergyPrice _{ojt}	Price of bioenergy product o at location j in period t (\$ per MWh)
BiofuelPrice _{nmt}	Price of biofuel n demanded by market m in period t (\$ per unit of product)
BiofuelTransportCost _{njmt}	Cost of transporting biofuel n from location j to market m in period t (\$ per unit of product)
BiomassCost _{fit}	Cost of purchasing biomass type f from source i in period t (\$ odmt ⁻¹)
BiomassTransportCost _{ijlt}	Cost of transporting biomass type f from source i to location j in period t (\$ odmt ⁻¹)
Yield _{lnf}	Yield of biofuel n per unit of biomass type f converted through technology l (units of product output per odmt of feedstock input)
Yield _{lof}	Yield of bioenergy product o per unit of biomass type f converted through technology l (units of product output per odmt of feedstock input)
FixedCost _{lt}	Fixed cost associated with technology l (\$) in period t
MaxCapacity _l	Maximum capacity of technology l (input biomass odmt.year ⁻¹)
MaxDemand _{nmt}	Maximum demand for biofuel n from market m in period t (units of product)
MaxDemand _{ojt}	Maximum demand for bioenergy product o at location j in period t (MWh of bioenergy)
MinDemand _{nmt}	Minimum demand for biofuel n from market m in period t (units of product)
MinDemand _{ojt}	Minimum demand for bioenergy product o at location j in period t (MWh of bioenergy)
VariableCost _{lt}	Variable cost associated with technology l in period t (feedstock dependent) (\$ odmt ⁻¹)
Decision variables	
q_{jlt}	Binary variable: 1, if technology l should be installed or is operating at location j in period t ; 0, otherwise
u_{fijt}	Amount of biomass type f from source i transported to location j in period t (odmt)
u_{fjlt}	Amount of biomass type f at location j used by technology l in period t (odmt)
v_{njlt}	Amount of biofuel n produced at location j with technology l in period t (units of product)
v_{ojlt}	Amount of bioenergy product o generated at location j with technology l in period t (units of product)
w_{njmt}	Amount of biofuel n transported from location j to market m in period t (units of product)
w_{ojt}	Amount of bioenergy product o sold at location j in period t (units of product)

^aOven-dry metric tonne (having 0% moisture content).

$$\begin{aligned}
 \text{TotalCost}_t = & \sum_{fi} (\text{BiomassCost}_{fit} * \sum_j u_{fijt}) \\
 & + \sum_{fij} (\text{BiomassTransportCost}_{fijt} * u_{fijt}) \\
 & + \sum_l (\text{FixedCost}_{lt} * \sum_j q_{jlt}) \\
 & + \sum_l (\text{VariableCost}_{lt} * \sum_{fj} u_{fjlt}) \\
 & + \sum_{nmj} (\text{BiofuelTransportCost}_{nmjt} * w_{nmjt})
 \end{aligned} \quad (3)$$

The constraint in Equation (4) restricts the amount of each biomass type f that can be purchased from each source i in each period t :

$$\sum_j u_{fijt} \leq \text{AvailableBiomass}_{fit} \quad \forall f \in F, i \in I, t \in T \quad (4)$$

To prevent the deterioration of biomass and the loss of dry matter, no biomass storage is considered in the model. The sum of biomass from all sources is fully utilized by all technologies in all periods (Eq. (5)).

$$\sum_i u_{fijt} = \sum_l u_{fjlt} \quad \forall f \in F, j \in J, t \in T \quad (5)$$

Inbound shipments of biomass type f to a conversion technology l at a location j can only occur when that technology at that location is operating in period t (denoted by the binary term q_{jlt}). The amount of biomass that can be sent to and used by the conversion technology in each period t is limited by its maximum capacity (Eq. (6)).

$$\sum_f u_{fjlt} \leq \text{MaxCapacity}_l * q_{jlt} \quad \forall j \in J, l \in L, t \in T \quad (6)$$

Equations (7) and (8) relate the amount of products generated by each technology l to the respective production yields.

$$v_{njlt} = \sum_f (\text{Yield}_{l,f} * u_{fjlt}) \quad \forall n \in N, j \in J, l \in L, t \in T \quad (7)$$

$$v_{ojlt} = \sum_f (\text{Yield}_{l,o} * u_{fjlt}) \quad \forall o \in O, j \in J, l \in L, t \in T \quad (8)$$

In each period t , the total amount of biofuel n that is distributed to markets and the total amount of bioenergy product o that is sold at a plant location cannot exceed the production levels (Eqs. (9) and (10)).

$$\sum_m w_{njmt} \leq \sum_l v_{njlt} \quad \forall j \in J, n \in N, t \in T \quad (9)$$

$$w_{ojt} \leq \sum_l v_{ojlt} \quad \forall o \in O, j \in J, t \in T \quad (10)$$

Equations (11) and (12) set the demand boundaries for bioenergy and biofuels.

$$\text{MinDemand}_{nmt} \leq \sum_j w_{njmt} \leq \text{MaxDemand}_{nmt} \quad \forall n \in N, m \in M, t \in T \quad (11)$$

$$\text{MinDemand}_{ojt} \leq w_{ojt} \leq \text{MaxDemand}_{ojt} \quad \forall o \in O, j \in J, t \in T \quad (12)$$

Equation (13) ensures that once a conversion technology l is installed at location j , it will continue operation in subsequent periods within the planning horizon.

$$q_{jlt} \geq q_{jlt-1} \quad \forall l \in L, j \in J, t \in T \quad (13)$$

In addition, non-negativity constraints and binary constraints are part of the model (Eqs. (14), (15), (16) and (17)).

$$u_{fjlt}, u_{fjlt} \geq 0 \quad \forall f \in F, i \in I, j \in J, l \in L, t \in T \quad (14)$$

$$v_{njlt}, v_{ojlt} \geq 0 \quad \forall j \in J, l \in L, n \in N, o \in O, t \in T \quad (15)$$

$$w_{njmt}, w_{ojt} \geq 0 \quad \forall j \in J, m \in M, n \in N, o \in O, t \in T \quad (16)$$

$$q_{jlt} \in \{0, 1\} \quad \forall l \in L, j \in J, t \in T \quad (17)$$

4. CASE STUDY

To demonstrate its applicability, the developed model was applied to a case study in the Williams Lake TSA located in the Interior Region of British Columbia, Canada. This TSA covers around 4.9 million hectares and is one of the largest TSAs in the province [42]. It has been significantly affected by the mountain pine beetle infestation, and almost all harvesting in the area has been focused on salvage of beetle-killed lodgepole pine trees. The harvest

has been used mainly to fulfill the demand of sawmills located in three communities: Anahim Lake, Hanceville and of Williams Lake. Currently, the majority of harvesting residues (i.e., tree tops, branches, leaves and non-merchantable stems) are left in the forest and burned, and a large amount of sawmill wastes (i.e., sawdust, shavings, chips and hog fuel) is burned or landfilled.

This case study examines different utilization alternatives for forest residues that are not committed to any other industry in the region. The production of four products is considered: heat, electricity, bio-oil and pellets. The communities of Anahim Lake, Hanceville and Williams Lake are the potential locations of new plants. The MILP model selects the best technologies at each location (specifying the year when each installation should be carried out) and determines the optimal annual flows of biomass residues and products for each of the 20 one-year periods.

All cost figures in this case study are adjusted for currency exchange and inflation rate to 2013 CAD\$ (Canadian currency), and a 5% depreciation rate is used in the model for depreciable assets.

4.1. Forest residues

Two sources of forest residues are considered: (i) sawmill wastes available at each location in the form of wood chips and hog fuel (average annual amounts and cost estimates obtained from [43,44]), and (ii) harvesting residues collected from 1592 aggregated cutting blocks throughout the Williams Lake TSA in the form of chips from non-merchantable mountain pine beetle stems, and hog fuel from harvesting residues (amounts and cost estimates per period supplied by FPIInnovations). Forest residue types and their characteristics are described in Table II.

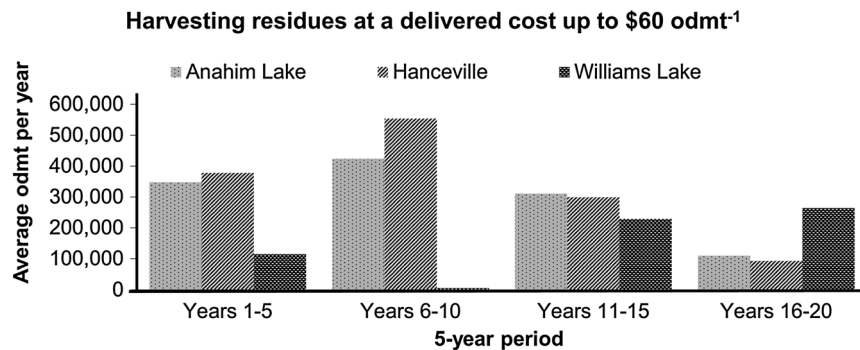
Based on the 20-year harvesting block data of the TSA, the amount of harvesting residues that can be economically available for bioenergy or biofuels at each of these communities is expected to change over the next 20 years. Figure 2 shows the estimated annual amounts of harvesting residues that can be procured at a delivered cost of up to \$60 odmt⁻¹ (based on the data supplied by FPIInnovations). Note that the harvesting block data are available in 5-year periods and the average annual availability of each aggregated cutting block was obtained by dividing its 5-year period's total amount by five. In our analysis, additional harvesting residues volumes are available at each location for delivered costs greater than \$60 odmt⁻¹.

4.2. Biomass transportation

Harvesting residues are chipped at the roadside and transported to the plant. Delivered costs of harvesting residues of each type from each aggregated cutting block to each potential location were taken from the output of FPIInterface, a forest operations model developed by FPIInnovations [49]. We assumed that sawmilling wastes are free of charge if utilized at the location where they were produced, and if used by a plant in a different location,

Table II. Forest residues types, ash and energy contents, and bio-oil yield.

Type		Ash content (%)	Energy content (GJ odmt ⁻¹)	Bio-oil yield (m ³ odmt ⁻¹)
Sawmill wastes	Wood chips	0.52 ^a	17.72 ^a	0.656 ^e
	Hog fuel	4.75 ^a	17.80 ^a	0.492 ^e
Harvesting residues	Chips from mountain pine beetle stems	1.00 ^b	19.00 ^d	0.643 ^e
	Hog fuel from harvesting residues	1.92 ^c	19.00 ^c	0.607 ^e

^aDerived from [45].^b[46].^cSample W6 in [14].^dAverage from case 3 in [47].^eDerived from [48] with basis on ash content.**Figure 2.** Harvesting residues (including chips from mountain pine beetle stems and hog fuel) at a delivered cost up to \$60 odmt⁻¹ for each location.

\$25.00 per odmt for wood chips and \$10.00 per odmt for hog fuels are charged. Transportation of sawmill wastes is carried out using a tri-axle walking floor trailer with a weight capacity of 28 metric tonnes. Transportation cost is assumed to be \$126.5 h⁻¹, with average transportation speeds of 86 (empty) and 77 km h⁻¹ (loaded) [44].

4.3. Biomass conversion

Four types of biomass conversion technologies are considered: (i) heat and/or electricity generation through combustion; (ii) heat and/or electricity generation through gasification; (iii) production of pellets; and (iv) production of bio-oil through fast pyrolysis. Table III describes the biomass feedstock requirements for the considered technologies.

Combustion and fast pyrolysis processes can convert all types of forest residues presented in Table II; however,

gasification and pelletization processes require clean and low ash content feedstock. Thus, for gasification and pelletization technologies, only two feedstock types have been included: wood chips (sawmilling waste) and chips from mountain pine beetle stems (harvesting residues). Tables IV and V show the relevant technology-related input parameters. In Tables IV and V, fixed cost parameters include annualized capital and maintenance costs of preparation yards, storage facilities, preprocessing (drying and hammer milling) and conversion technologies; variable costs include annual operating and labor costs related to each technology type and size. Costs of heat and power producing technologies are based on Marinescu [43,55], whose estimations are derived from published reports [12,50], and communication with relevant technology suppliers (Table IV). In Table IV, efficiency refers to the net energy efficiency, that is, the ratio (%) of the energy output (heat or electricity)

Table III. Feedstock requirements for selected conversion technologies.

Conversion process	Type of forest residues	Ash content	Moisture content	Particle size
Combustion (fixed bed)	All types ^a	<30% ^b	10–50% ^a	5–50 mm ^a
Gasification (fixed bed)	Chipped wood ^a	<6% ^b	<20% ^c	<75 mm ^a
Pelletization	Chipped wood ^e	<1% ^e	<10% ^d	<6 mm ^d
Pyrolysis (fluidized bed)	All types ^f	Unknown ^f	<10% ^f	<2 mm ^f

^a[50].^b[45].^c[51,52].^d[53].^e[54].^f[48].

Table IV. Conversion efficiencies, fixed and variable costs of considered bioenergy facilities (based on Marinescu [43,55]).

Processing facility type	Capacity	Efficiency		Fixed cost per year (\$)	Variable cost ^a (\$odmt ⁻¹ of biomass)
		Thermal (%)	Electrical (%)		
Biomass boiler (heat only)	2 MW _{th}	71.3	0.0	676,637	47.3
Biomass boiler + steam turbine (CHP ^b)	0.5 MW _{el}	21.7	14.5	336,152	101.4
	2 MW _{el}	20.2	16.2	1,228,730	38.0
	3 MW _{el}	14.1	16.9	1,738,419	30.2
	5 MW _{el}	12.9	18.3	2,664,946	23.4
Biomass boiler + steam turbine (electricity only)	0.5 MW _{el}	0.0	18.7	336,152	130.8
	2 MW _{el}	0.0	18.7	1,228,730	43.9
	3 MW _{el}	0.0	18.7	1,738,419	33.4
	5 MW _{el}	0.0	18.7	2,664,946	23.9
Biomass oil heater + ORC ^c (CHP or electricity only)	0.5 MW _{el}	48.5	11.1	693,312	89.0
	2 MW _{el}	52.1	12.0	1,660,422	31.9
	3 MW _{el}	54.1	11.5	2,585,966	25.8
	5 MW _{el}	55.5	11.8	4,233,396	21.1
Biomass gasifier + ICE ^d (CHP or electricity only)	0.5 MW _{el}	10.3	12.6	284,640	164.7
	2 MW _{el}	9.0	19.2	1,051,479	82.3
	3 MW _{el}	9.0	18.7	1,493,903	61.1
	5 MW _{el}	8.7	21.2	3,065,430	48.7

^aVariable cost was assumed to be the same for all periods.^bCombined heat and power.^cOrganic rankine cycle.^dInternal combustion engine.**Table V.** Yield, fixed and variable costs of considered pyrolysis and pellet plants.

Processing facility type	Capacity	Yield	Fixed cost per year (\$)	Variable cost ^a (\$odmt ⁻¹ of biomass)
Pyrolysis	200 odmt day ⁻¹	Feedstock dependant ^b	2,653,467	44.1
	400 odmt day ⁻¹	Feedstock dependant ^b	4,940,351	38.3
	600 odmt day ⁻¹	Feedstock dependant ^b	7,058,729	37.0
Pellets	15,000 t year ⁻¹	91% ^c	392,620	39.0
	30,000 t year ⁻¹	91% ^c	874,774	50.3
	45,000 t year ⁻¹	91% ^c	1,089,497	43.1

^aVariable cost was assumed to be the same for all periods.^bRefer to Table II.^cDerived from [57].

divided by the energy contained in the input biomass fuel. Costs of pellet producing technologies are based on [46,53,56], and costs of pyrolysis technologies are based on [46,48] (Table V).

4.4. Product transportation and distribution

It was assumed that the heat generated will meet local demand, electricity will be sold to the grid, and pyrolysis bio-oil and pellets (regarded as biofuels in this study) will be transported to markets or ports. The market for pyrolysis bio-oil is Williams Lake, which is the largest community in the region. The market for pellets is the pellet export facility located in the port of North Vancouver (rail distance of 505 km from Williams Lake). The transportation of pyrolysis bio-oil is carried out using semi-trailer tanks with five axles and a volume capacity of 27,276 L. The cost of transporting a full truck load of bio-oil is assumed to be \$2.02 km⁻¹ [58]. Pellets transportation has

two stages: first, by truck to Williams Lake, then by rail to the port of North Vancouver. The truck transportation of pellets is assumed to be outsourced to the same trucking company transporting the biomass, and the rail transportation to the port costs \$29 t⁻¹ [59].

4.5. Demand

The demand for heat and electricity in each location is estimated using previous technical reports and literature, and is assumed stable over the planning horizon. The demand for pellets and bio-oil was based on the total economically available forest residues in the region. The price of heat and electricity is based on current energy prices in each community (i.e., Anahim Lake is not connected to the Hydro grid and produces electricity from diesel at a price of \$0.285 kWh⁻¹), the price of bio-oil is derived from the current cost of household heating oil in the region (a liter

of bio-oil contains 54% of the energy content of heating oil) and the price of pellets is based on the current price of pellets for export at the port of North Vancouver (Table VI).

5. RESULTS

The MILP model was solved using the AIMMS 3.11TM software that integrates modeling language, a graphical user interface and numerical solver [60]. A set of constraints was added that forced the model to select, at each location, just one technology type and size for each type of bioenergy or biofuel product. The model for this case study had 136,961 constraints and 395,881 decision variables of which 1380 were integers. With an Intel 2.67-GHz processor, an optimal solution was found in 28 min, after 1,022,243 iterations.

5.1. Base case scenario

Table VII shows the optimum conversion technology types and capacities at each location determined by the model for the base case scenario. The installation of a biomass boiler connected to a steam turbine was selected as the best alternative for electricity production because of low or no demand for heat in Anahim Lake and Hanceville. In Williams Lake, the model selected a biomass boiler that was the heat generation technology with the highest thermal efficiency. The introduction of fast pyrolysis plants in the region seemed promising if there was enough demand for pyrolysis bio-oil. The location and size of pyrolysis plants determined by the model were based on the availability of residues around each location, and the closeness of each location to the market for bio-oil (Williams Lake). For all technologies, the recommended installation period was at the beginning of year 1, except for the pyrolysis plant in Williams Lake where installation was recommended at the beginning of year 11. The production of pellets was not recommended in any location by the optimal solution.

The NPV of the optimal solution for the 20-year planning horizon was \$565,488,983. The total discounted revenue of the supply chain considering all the selected technologies was \$827,078,521, and the total discounted cost was \$261,589,538. Bio-oil, electricity and heat sales accounted for about 81.5%, 18.2% and 0.3% of the revenue, respectively. Because the demand and price of products were assumed to be fixed over the planning period, revenue remained the same over the 20-year period. However, because of fluctuations in availability of harvest residues over time (changes occur in each 5-year period), the total cost components (sum of all selected plants) fluctuated as well (Figure 3). The largest fluctuations happen at the beginning of year 11, with the installation of a 200 odmt day⁻¹ pyrolysis plant. In year 11, bio-oil transportation cost decreased by 47%, biomass procurement cost decreased by 7%, annual fixed cost increased by 33% and annual variable costs increased by 9% compared with those in year 10.

5.2. Mountain pine beetle pessimistic scenario

In the base case scenario, it was assumed that because of the mountain pine beetle infestation, only 70% (by weight) of the lodgepole pine logs harvested and transported to roadside was merchantable, 15% was left in the forest and the remaining

Table VII. Optimal selection of technologies for each location—base case scenario.

Location	Technology	Capacity
Anahim Lake	Biomass boiler + steam turbine electricity only	5 MW
Hanceville	Biomass boiler + steam turbine electricity only	0.5 MW
Hanceville	Pyrolysis plant	400 odmt day ⁻¹
Williams Lake	Biomass boiler	2 MW
Williams Lake	Pyrolysis plant	200 odmt day ⁻¹

Table VI. Product demand and price parameters (parameters assumed the same for all periods).

Products	Demand (min–max) at location/market				Units
	Anahim Lake	Hanceville	Williams Lake	Port of export	
Heat	0–19,710 ^a	–	12,322–12,322 ^c	–	MWh
Electricity	3942–39,420 ^a	3942–39,420 ^b	–	–	MWh
Bio-oil	–	–	0–81,377,049	–	l
Pellets	–	–	–	0–135,000	t
Products	Price of products at location/market				Units
	Anahim Lake	Hanceville	Williams Lake	Port of export	
Heat	0.095	–	0.014	–	\$ MWh ⁻¹
Electricity	0.285	0.095	–	–	\$ MWh ⁻¹
Bio-oil	–	–	0.64	–	\$ l ⁻¹
Pellets	–	–	–	142.5	\$ t ⁻¹

^a[55].

^b[43].

^c[44].

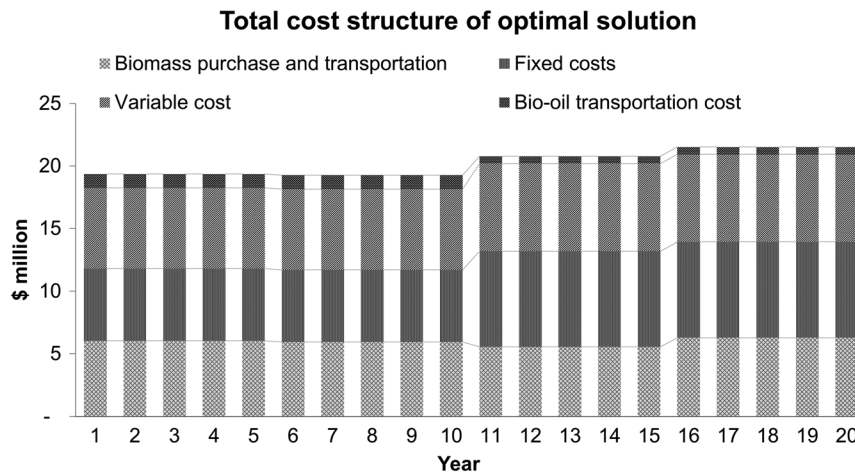


Figure 3. Breakdown of cost components over the planning horizon.

15% was available for biofuels or bioenergy (chips from mountain pine beetle stems in Table II) [61]. As time passes, the quality of biomass from MPB trees deteriorates; therefore, this scenario was developed in which no chips from mountain pine beetle stems were available for energy after year 10. No significant changes in the configuration of the supply chain were noticed because in the original optimal solution, the use of mountain pine beetle logs was not significant. The total biomass transportation cost slightly increased by 3% as a result of longer transportation distances to fulfill the required biomass supply.

5.3. Sensitivity analysis

To evaluate the effect of potential variations in model parameters on the supply chain profit, a sensitivity analysis was performed using the base case scenario. The variation was considered in these parameters: (i) forest residue cost; (ii) forest residue availability; (iii) product demand; and (iv) product (bioenergy and biofuels) price. Changes in all these parameters were selected based on possible biomass supply and energy market conditions in the area. All the evaluated cases resulted in feasible solutions.

The sensitivity of the model to forest residue cost was analyzed separately for harvesting residues and for sawmill wastes. The cost of harvesting residues was increased from \$0 to \$20 odm^{-1} , and the cost of sawmill wastes (both hog fuel and wood chips) was increased from \$0 to \$25 odm^{-1} for hog fuel and to \$50 odm^{-1} for wood chips. Product prices were varied one at a time. Heat and electricity costs were varied from \$0.095 to \$0.285 kWh^{-1} independently for each location to evaluate the effect of changes in the price of electricity and the cost of heat in each community, except for Williams Lake, where biomass based heating was only compared with the current cost of natural gas heating (the base case scenario). The price of pellets was changed from \$135 to \$150 t^{-1} to capture the price variability of such product, and bio-oil price was varied by $\pm 20\%$. Similarly, the maximum demand for products was evaluated individually for each type of product that was deemed relevant. The demand for bio-oil was changed by $\pm 20\%$, the demand for pellets was increased by 20% and the demand for heat in Anahim Lake was changed by $\pm 20\%$. Finally, the biomass availability for all sources was changed by $\pm 20\%$ from the base case scenario. Figure 4 summarizes the results of the sensitivity analysis.

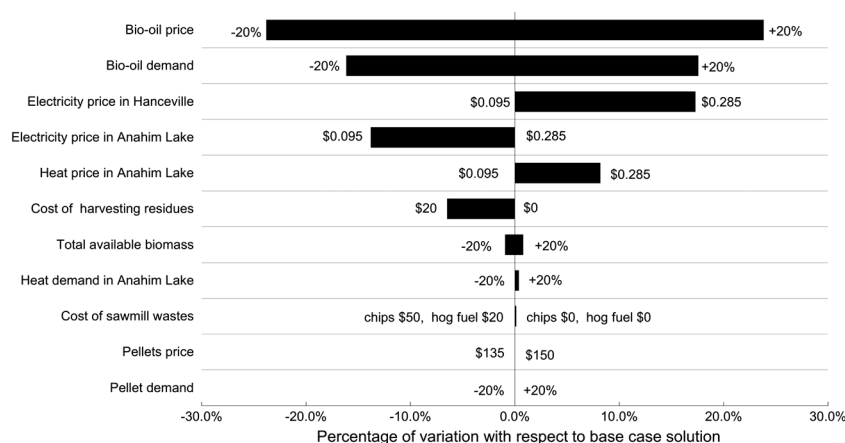


Figure 4. Sensitivity of optimal solution to selected parameters.

Table VIII. Impact of changes in parameters on technology selection.

Changes in parameters								
Location	Conversion technology and size	Bio-oil demand decreased by 20%	Bio-oil demand increased by 20%	Bio-oil price decreased by 20%	Electricity price in Hanceville increased from \$0.095 to \$0.190 or \$0.285 kWh ⁻¹	Electricity price in Anahim Lake reduced from \$0.285 to \$0.095 kWh ⁻¹	Heat demand in Anahim Lake increased by 20%. (OR) Heat price in Anahim Lake increased from \$0.095 to \$0.190 or \$0.285 kWh ⁻¹	Sawmill waste price decreased to \$0.00 odmt ⁻¹ . (OR) Available biomass increased by 20%
Anahim Lake	<i>Biomass boiler + steam turbine electricity only (5 MW)</i>	1	1	1	1			1
	Biomass boiler + steam turbine CHP ^a (3 MW)					1		
	Biomass oil heater + ORC ^b (5 MW)						1	
	Biomass boiler + steam turbine electricity only (5 MW)				1			1
Hanceville	<i>Biomass boiler + steam turbine electricity only (0.5 MW)</i>	1	1	1				
	<i>Pyrolysis plant (400 odmt day⁻¹)</i>	1	1	1	1	1	1	1
	Pyrolysis plant (600 odmt day ⁻¹)							
	<i>Biomass boiler (2 MW)</i>	1	1	1	1	1	1	1
Williams Lake	<i>Pyrolysis plant (200 odmt day⁻¹)</i>			16	11	11	11	16

^aCombined heat and power.^bOrganic rankine cycle.

Table VIII shows those changes in the parameters that affected the technology selection. Technologies selected in the base case scenario are in bold and italics. Numbers in cells represent the periods when the installation was recommended for each case.

The impact of variation in some parameters was high on the total NPV (Figure 4) and also changed the decision on technology selection (Table VIII). When the bio-oil price decreased by 20%, the same technologies were chosen, but the installation of the 200 odmt year⁻¹ pyrolysis plant in Williams Lake was recommended in year 16 (when there are more available amounts of low-cost harvesting residues around Williams Lake) instead of year 11 (in the base case solution), with a loss in profit around 24%. When the bio-oil demand decreased by 20%, the installation of a pyrolysis plant in Williams Lake was not recommended, and the profit loss was around 16%. On the other hand, when the bio-oil demand increased by 20%, the total production of bio-oil was concentrated in Hanceville (instead of having two pyrolysis plants, one in Hanceville and another one in Williams Lake, the model chose the largest size to be installed in Hanceville), and the gain in profit was around 18%. Other significant parameters were the price of electricity in Hanceville and Anahim Lake, the heat price in Anahim Lake, the cost of harvesting residues and the total available biomass as shown in Figure 4.

6. DISCUSSION

Because of the case-specific nature of supply chain optimization problems, we cannot compare the results of the supply chain designed in this study with those of previous biofuel and bioenergy supply chain optimization models. However, in our discussion, we can refer to the recommendations of the Biopathways Project, which performed techno-economic comparisons among different forest biomass products in the interior region of British Columbia [10].

Overall, the solution suggested by the model was consistent with the results of the Biopathways Project [10] in which the installation of fast pyrolysis plants in the region was suggested as the most profitable alternative (pyrolysis bio-oil sales accounted for 81% of the revenue in our base case solution). However, it is worth noting that biomass fast pyrolysis technologies and their products are not commercial yet (in our study, bio-oil price was derived based on the price of heating oil, and a high demand for bio-oil was assumed). Currently, there are pilot demonstration plants (e.g., Union Fenosa in Spain, Dynamotive in Canada and Wellman in the UK), and some companies are promoting the commercialization of bio-oil in the world (e.g., Dynamotive and Ensyn Technologies) [62,63].

While the Biopathways Project identified CHP via gasification as a promising process for Interior British Columbia [10], the results of our model did not support the selection of CHP systems because of the current low heat demand and price in communities that demand electricity (Anahim Lake and Hanceville). In the base case

scenario, electricity-only production in biomass boilers coupled with steam turbines was selected in Anahim Lake and Hanceville. The reason for this is the higher electrical efficiency of such technology. In the sensitivity analysis, when heat demand or price in Anahim Lake was increased, the model selected biomass boilers coupled with organic rankine cycle systems instead of gasification because of their higher thermal efficiency. Selected combustion technologies (biomass boilers coupled to steam turbines or to organic rankine cycles) are commercially available; however, communities would require large capital investments for their installation.

Current heat and electricity production costs are different for each of the three communities in the study. Anahim Lake has the largest costs because of a complete dependence on diesel generation (at a cost of \$0.285 kWh⁻¹). Our results suggest that the combustion of forest biomass could significantly reduce electricity costs in Anahim Lake. On the contrary, in the community of Hanceville, electricity production from forest biomass would be suitable if sawmill residues could be obtained from suppliers at no cost or at higher electricity prices (e.g., above \$0.190 kWh⁻¹).

The multi-period nature of the developed model allowed us to capture fluctuations in harvesting residue availability in the Williams Lake TSA, which influenced the long-term strategic planning of technology investment decisions. In this study, product demand and price were considered steady over the planning horizon. However, the sensitivity analysis showed that uncertainty in those parameters affects the optimal solution (especially for bio-oil). The demand for electricity in British Columbia is expected to grow by 45% during the next 20 years [64] (in our case study we evaluated changes in electricity demand of up to 20% since the communities under analysis have small population) and the electricity price is expected to increase by more than 20% from 2013 levels by 2016 [65]. In addition, as bio-oil applications develop over time, bio-oil price fluctuation might occur [66]. Our future work will involve complete modeling of uncertainty in relevant parameters. Also, future work will entail the evaluation of the associated environmental and social impacts of each process, and the incorporation of those aspects within the design of the forest residue supply chain for bioenergy and biofuel production.

7. CONCLUSIONS

The competitive use of forest residues as an alternative raw material for the production of bioenergy and biofuels requires an optimal allocation of raw material and capital resources. The model presented in this study aids strategic decisions on technology investment and annual material flows in the forest residues to bioenergy and biofuel supply chain. The model presented here has two advantages over previous approaches. First, it considers the integrated production of biofuel and bioenergy from different types of forest residues (harvesting residues and mill wastes). Second, the multi-period setting of the model allows the

consideration of supply and market parameters' variability over time in making strategic decisions.

We demonstrated the appropriateness of our approach through a case study in three communities located in British Columbia. Based on the region's timber harvesting plans for the next 20 years, a net present value of \$565,488,983 could be generated by producing bio-oil, heat and power from available harvesting residues and sawmill wastes.

The effect of variations in the parameters of the model was assessed through a sensitivity analysis. The results of the sensitivity analysis demonstrated that the most important parameters that affect the net present value of the supply chain were bio-oil price and demand. The demand for heat and electricity and the price that each market was willing to pay for them played a significant role in deciding which bioenergy technology to install in each location. An extension of the model could include uncertainties in the demand and price of each product. Our future work will involve a complete modeling of uncertainty in relevant parameters.

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